1 TestTranslation

1.1 System overview

TestTranslation demonstrates and tests translation bridges of the TranslationManager. It is an interface translator and a data model translator in one native Go executable (`ah5-support-go`, role `TestTranslation`):

- Interface translator: the TranslationManager asks it which targets it can reach (`check-targets`) and has it create bridges (`initialize-bridge`) and remove them (`abort-bridge`). A bridge forwards the consumer's calls of one operation to the target, translates payloads with data model translators and reports usage to the TranslationManager (`translationReport`).
- Data model translator: translates between `simpleJson` (`[{name, value|stringValue|boolValue, unit, time}]`) and `senml` (RFC 8428 records; base name, time and unit resolved).

1.2 Produced and consumed services

Produced: `interfaceBridgeManagement` (metadata `interfaceBridge: {from: [generic_https], to: generic_https}`), `dataModelTranslation` (metadata `dataModelIds: [[simpleJson, senml], [senml, simpleJson]]`), `echo`, `generalManagement`, and native `sysMonitor` telemetry. Each bridge is offered to its consumer with policy `TRANSLATION_BRIDGE_TOKEN_AUTH`; bridges are not registered services. Consumed: the bridge targets (any system of the local cloud), data model translators (its own `dataModelTranslation`), the TranslationManager's `translationReport`, and the ServiceRegistry, Authentication and ConsumerAuthorization authorities.

1.3 Important delimitations

TestTranslation is a test application system for the TranslationManager, not a production translator.

- One bridge kind: `generic_https` to `generic_https`, one negotiated operation per bridge; targets with `CERT_AUTH` or `NONE` policies (authorization tokens for token-protected targets are not presented).
- Data models `simpleJson` and `senml` only; translations are synchronous (a task is `DONE` or `ERROR` when `init-translation` returns).
- Bridges are kept in the system's JSON state; translation tasks only in memory (latest 1000).

1.4 Security and persistence

Mutual TLS with local-cloud system certificates. The management services require an identity token bound to the client certificate and a ConsumerAuthorization grant (TranslationManager for `interfaceBridgeManagement`, the interface translator for `dataModelTranslation`). A bridge requires a local-cloud certificate and `Authorization: Bearer TRANSLATION-BRIDGE-TOKEN//<bridgeId>`; only the negotiated method and path template are forwarded. Bridge targets and data model translators are called with TestTranslation's identity; their certificates are verified against the cloud CA, the `sy` profile, the cloud name and the host name (the Java bridge payloads carry no provider name to pin).

2 References

Contracts: Java `TranslationManager` drivers (`InterfaceTranslatorDriver`, `DataModelTranslatorFactoryDriver`) and the common DTOs `TranslationCheckTargetsRequestDTO`, `Tr`

anslationBridgeInitializationRequestDTO, DataModelTranslationInitRequestDTO, DataModelTranslationResultResponseDTO. Source: internal/core/testtranslation.go; test: scripts/smoke-translation.sh of Arrowhead-test-cloud-go.